

Assignment

1. Explain Malware

Malware is short for "malicious software." It refers to any software or program that is designed to harm, disrupt, steal data, or gain unauthorized access to a computer system, network, or device. Common types of malware include viruses, worms, ransomware, spyware, and trojans. Cybercriminals use malware to steal sensitive information, damage files, monitor user activities, or demand money from victims.

2. Identify Indicators

Indicators of malware infection are signs that a system may have been compromised. Some common indicators include:

- Slow computer performance without any clear reason.
- Frequent system crashes or unexpected restarts.
- Unusual pop-up messages or advertisements.
- Unknown programs running on the device.
- Missing, modified, or encrypted files.
- Increased network activity when the device is not being used.
- Disabled security software such as antivirus programs.
- Unauthorized access to accounts or suspicious login activity.

These indicators help users and security teams detect potential malware infections early.

3. Describe Response

When malware is detected, an appropriate response should be taken to minimize damage and restore security. The response process includes:

1. **Identify the Threat** – Confirm that malware is present and determine its type.
2. **Isolate the System** – Disconnect the affected device from the network to prevent the malware from spreading.
3. **Remove the Malware** – Use antivirus or anti-malware tools to scan and eliminate the malicious software.
4. **Recover Data and Systems** – Restore affected files from backups and repair any damaged systems.
5. **Update Security Measures** – Install software updates, change passwords, and strengthen security controls.

6. **Monitor and Review** – Continue monitoring the system to ensure the threat has been completely removed and to prevent future incidents.

In conclusion, understanding malware, recognizing its indicators, and responding effectively are important steps in maintaining cybersecurity and protecting digital assets.